Theory of Elementary Particles: Lecture 2

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- Derivation
- State normalization in the finite volume

Cross-section in QFT

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Scalar Field

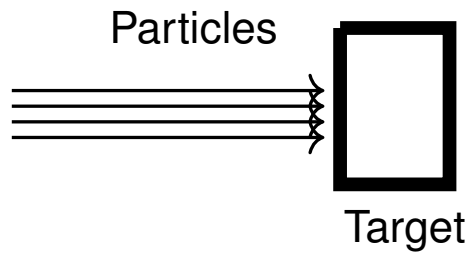
Fermion field

Feynman rules for $i\mathcal{M}$ in QED

Cross-Section Formula

Preliminaries

cross section (σ) = effective size of the target particle.

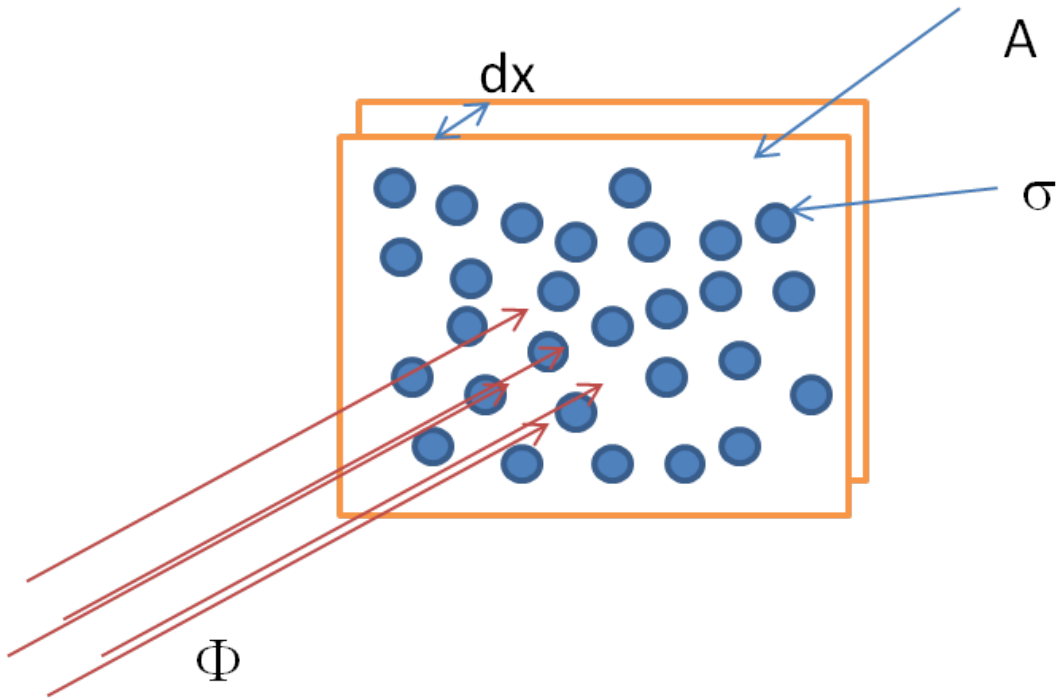


σ is measured in barns

$$\begin{aligned} 1\,\text{barn} &= 10^{-28}\,\text{m}^2 \\ &= (10\,\text{fm})^2 \\ &= 100\,\text{fm}^2 \end{aligned}$$

Remark

Nucleon has a radius of the order of 1 fm, hence, the effective nucleon area is about 1 fm$^2 = 0.01$ barn!



- Let us consider the beam of particles moving along x axis, which collide with the thin layer (perpendicular to the x axis) of width dx and of area A.
- The cross section σ can be understood as the effective area of the particle.

- Let us introduce the flux Φ of particles defined in unit area and time

$$\Phi\left[\frac{1}{s\cdot m^2}\right] = \rho v \leftarrow \frac{N_{tot}}{dxA}\frac{\Delta x}{\Delta t} \rightarrow \frac{N_{tot}}{\Delta x A}\frac{\Delta x}{\Delta t} = \frac{N_{tot}}{A\Delta t} \tag{1}$$

- Notice that here $dx \rightarrow \Delta x$
- ρ is density of particles in the beam, v-velocity of particles
- N_{tot} number of particles crossing targets

- Assume that in the unit volume of the target there are n scattering centers, which do not overlap
- The chance for scattering of particle from the beam with the target reads

$$P = \frac{\sigma \cdot nAdx}{A} = \sigma n dx = \sigma n', \tag{2}$$

where n' is the number of centers in the unit area

- Let $\Phi_0 \equiv \Phi(x=0)$ denotes the initial flux of particles
- The total number of incoming particles (in the unit time) reads

$$\frac{N_{tot}}{\Delta t} = \Phi_0 \cdot A \tag{3}$$

- The number of particles which collide in short period of time and then disappear from the beam reads

$$d\Phi = -P\Phi = -\Phi \frac{nA\sigma dx}{A} \Rightarrow \Phi(x) = \Phi_0 \exp\left[-n\sigma x\right]. \tag{4}$$

- The number of the scattered particles in the unit time and the unit area reads

$$N_{sct} = \underbrace{\Phi_0}_{total} - \underbrace{\Phi_0 \exp\left[-n\sigma x\right]}_{after\ scat.} \approx \Phi_0 - \Phi_0(1-n\sigma x) = \Phi_0 n\sigma x \tag{5}$$

- N_{sct} - number of scattered particles in unit of time

- Then the cross-section is given by

$$\sigma = \frac{N_{sct}}{\Phi_0 n x} = \frac{N_{sct}}{\Phi_0 n'}. \tag{6}$$

- ratio

$$\tilde{P} = \frac{N_{sct}}{n'} = [\text{the number of scattering events per target per unit time}] = \frac{P}{T} \tag{7}$$

is the probability of the scattering that occurs in given unit area at given short moment of time assuming that a particle from the beam interacts with only one scattering center in the layer at the given, short, moment of time.

Cross-section formula

Eventually we get the formula for the cross section,

$$\sigma = \frac{P}{T\Phi_0}. \tag{8}$$

The Φ_0 is a number of beam particles crossing the area A in a unit time. Then if one introduces

- ρ density of particles;
- v a relative velocity between beam and target.

we easily get the formula

$$\Phi = \rho v = \rho \times \frac{dX}{dT} = \frac{\#\text{of particles per area}}{\text{unit time}}.$$

In QFT one assumes that,

$$\rho = \frac{1}{V}, \quad \Phi = \frac{v}{V},$$

where V is the volume of the system.

The cross section formula for QFT

$$\sigma = V \frac{P}{Tv}. \tag{9}$$

Our goal is to obtain probability P from QFT or QM

Remark

We need to calculate from QFT probability $P \sim |\langle final|S|initial\rangle|^2$, but in infinite volume there are some problems with proper definition of P..., we shall consider QFT in finite volume...

- **non-relativistic quantum mechanics:** in the finite volume one can properly normalize the wave function of the freely moving particle and relates the S matrix elements with the probabilities.
- our idea is to derive the cross section formulas in the finite volume and then to take the infinite volume limit

The momentum eigenfunction equation for freely moving particle reads

$$-i\nabla u_{\mathbf{p}}(\mathbf{r}) = \mathbf{p}u(\mathbf{r}) \tag{10}$$

one finds solution of the above equation

$$u_{\mathbf{p}} \sim \exp(i\mathbf{p}\cdot\mathbf{r}). \tag{11}$$

If we assume the periodic boundary condition

$$u_{\mathbf{p}}(\mathbf{r}) = u_{\mathbf{p}}(\mathbf{r} + \mathbf{L}_k), \text{ for } k = x, y, z, \ \mathbf{L}_k = L(\delta_{xk}, \delta_{yk}, \delta_{zk}). \tag{12}$$

then normalized particle wave function is given by

$$u_{n_x,n_y,n_z}(\mathbf{r}) = \frac{1}{\sqrt{L^3}} \exp\left(2\pi i \frac{\mathbf{n}\cdot\mathbf{r}}{L}\right), \quad \mathbf{n} = (n_x, n_y, n_z). \tag{13}$$

the momentum is quantized and it takes the discrete values

$$\mathbf{p} = \frac{2\pi}{L}(n_x, n_y, n_z), \quad V = L^3 \tag{14}$$

it is easy to see that,

$$\int_V d^3r \, u^*_{n_x,n_y,n_z}(\mathbf{r}) u_{m_x,m_y,m_z}(\mathbf{r}) = \delta_{n_x m_x}\delta_{n_y m_y}\delta_{n_z m_z} \tag{15}$$

Relation between Dirac and Kronecker deltas

$$\lim_{L\to\infty}\int_{-L/2}^{L/2} e^{-ix(p'_x-p_x)}dx = 2\pi\delta(p'_x - p_x).$$

on the other hand

$$\int_{-L/2}^{L/2} dx e^{-ix(p'_x-p_x)} = L\frac{\sin\left(x\frac{2\pi(n'_x-n_x)}{L}\right)}{2\pi(n'_x-n_x)}\Bigg|_{-L/2}^{L/2} \quad (16)$$

$$= L\frac{\sin\left(\pi(n'_x-n_x)\right)}{\pi(n'_x-n_x)} = L\delta_{n'_x,n_x} = L\delta_{p'_x,p_x}. \quad (17)$$

so we have

$$\delta(p'_x - p_x) \to \frac{L}{2\pi}\delta_{p'_x,p_x} \quad (18)$$

As the result we introduce the Dirac delta in the finite volume V as it follows

$$\delta_V^{(3)}(\mathbf{p}' - \mathbf{p}) = \frac{V}{(2\pi)^3}\delta_{\mathbf{p}',\mathbf{p}}. \tag{19}$$

It should satisfy the property

$$\lim_{V\to\infty} \delta_V^{(3)}(\mathbf{p}' - \mathbf{p}) = \delta^{(3)}(\mathbf{p}' - \mathbf{p}) \tag{20}$$

The density of states in the finite volume reads

$$d^3p = \frac{(2\pi)^3}{V} d^3n \tag{21}$$

We keep the following one-particle state normalization

$$\langle \mathbf{p}' \mid \mathbf{p} \rangle = (2\pi)^3 2E_p \delta^{(3)}(\mathbf{p}' - \mathbf{p}) \tag{22}$$

then

$$\mid \mathbf{p} \rangle_{nor} = \frac{1}{\sqrt{(2\pi)^3 2E_p}} \mid \mathbf{p} \rangle, \quad {}_{nor}\langle \mathbf{p}' \mid \mathbf{p} \rangle_{nor} = \delta^{(3)}(\mathbf{p}' - \mathbf{p}) \to \delta_V^{(3)}(\mathbf{p}' - \mathbf{p}). \tag{23}$$

In the finite volume box-states are normalized so that,

$${}_{box}\langle \mathbf{p}' \mid \mathbf{p} \rangle_{box} = \delta_{\mathbf{p}',\mathbf{p}}$$

then

$$\mid \mathbf{p} \rangle_{box} = \sqrt{\frac{(2\pi)^3}{V}} \mid \mathbf{p} \rangle_{nor} = \frac{1}{\sqrt{2E_{\mathbf{p}} V}} \mid \mathbf{p} \rangle.$$

Cross-section in Quantum Field Theory

Physics

Let us consider the scattering process

$$A + B \to 1 + 2 + ... + N$$

The particle A collides with the particle B and as the result N particles are produced.

Goal: compute the probability dP. It can be expressed by the scattering matrix S_{fi}

$$\begin{aligned} dP(A, B \to 1, 2, ..., N) &\sim |_{out}\langle \mathbf{p}_1, \mathbf{p}_2, ...\mathbf{p}_N \mid A, B\rangle_{in}|^2 \\ &= |\langle \mathbf{p}_1, \mathbf{p}_2, ...\mathbf{p}_N \mid S \mid A, B\rangle|^2 \\ &= |S_{fi}|^2. \end{aligned}$$

- In the finite box $|S_{fi}^{box}|$ is the probability for interaction reads

$$P = |S_{fi}^{box}|^2 d^3n. \tag{24}$$

  where $d^3n = Vd^3p/(2\pi)^3$.
- For the N particles in the final state,

$$d^3n \to \prod_{i=1}^{N} \frac{Vd^3p_i}{(2\pi)^3} \tag{25}$$

- The probability density for the scattering reads,

$$P = |S_{fi}^{box}|^2 \prod_{i=1}^{N} \frac{Vd^3p}{(2\pi)^3}. \tag{26}$$

$$|S_{fi}^{box}|^2 = \frac{1}{V^{2+N}} \frac{1}{4E_A E_B} \prod_{j=1}^{N} \frac{1}{2E_{p_j}} |S_{fi}|^2 \tag{27}$$

then probability for the scattering reads

$$P = \frac{1}{2E_A 2E_B V^2} \prod_{i=1}^{N} \left(\frac{d^3 p_i}{2E_{p_i} (2\pi)^3} \right) |S_{fi}|^2. \tag{28}$$

S matrix contains a trivial part (non-interaction) and interacting part described by the T matrix.

$$S = 1 + iT. \tag{29}$$

then the probability for scattering reads

$$P = \frac{1}{4E_A E_B V^2} \prod_{i=1}^{N} \left(\frac{d^3 p_i}{2E_{p_i}(2\pi)^3} \right) |T_{fi}|^2. \tag{30}$$

When one applies (9) then one gets

$$d\sigma = \frac{1}{4E_A E_B V T v} \prod_{i=1}^{N} \left(\frac{d^3 p_i}{2E_{p_i}(2\pi)^3} \right) |T_{fi}|^2. \tag{31}$$

In most of the cases T matrix can be written as,

$$iT_{fi} = i(2\pi)^4 \delta^{(4)} \left(\sum_{i=1}^{N} p_i - p_A - p_B \right) \mathcal{M}_{fi}. \tag{32}$$

Unfortunately the Dirac delta in power of 2, in the $|T_{fi}|^2$,
hence one must consider first the Dirac delta in finite volume

$$\delta_V^{(4)}\left(\sum_{i=1}^{N} p_i - p_A - p_B\right)\delta_V^{(4)}\left(\sum_{i=1}^{N} p_i - p_A - p_B\right) = \delta_V^{(4)}\left(\sum_{i=1}^{N} p_i - p_A - p_B\right)\delta_V^{(4)}(0)\,, \tag{33}$$

where

$$\frac{VT}{(2\pi)^4} = \delta_V^{(4)}(0) \tag{34}$$

because

$$\delta_V^{(3)}(\mathbf{p}' - \mathbf{p}) = \frac{V}{(2\pi)^3}\delta_{\mathbf{p}',\mathbf{p}}. \tag{35}$$

Then the cross section reads,

$$d\sigma = \frac{1}{4E_A E_B v} \prod_{i=1}^{N} \left(\frac{d^3 p_i}{2E_{p_i} (2\pi)^3} \right) (2\pi)^4 \delta^{(4)} \left(\sum_{i=1}^{N} p_i - p_A - p_B \right) |\mathcal{M}_{fi}|^2. \quad (36)$$

v is the relative velocity of the beam particle with respect to the target, hence

$$v = \left| \frac{\mathbf{p}_A}{E_A} - \frac{\mathbf{p}_B}{E_B} \right| \quad (37)$$

in the laboratory frame

$$v = \left| \frac{\mathbf{p}_A}{E_A} \right| \tag{38}$$

it easy to see that

$$v = \frac{\sqrt{(p_A \cdot p_B)^2 - M_A^2 M_B^2}}{E_A E_B}. \tag{39}$$

Notice that the above derivation of the cross sections is valid only for co-linear particles momenta.

Decay Rate

Decaying of A particle into N final states:

$$A \to 1 + ... + N \tag{40}$$

$$d\Gamma = \frac{1}{2E_A} \prod_{f=1}^{N} \frac{d^3 p_f}{(2\pi)^3 2E_{p_f}} |\mathcal{M}_{fi}|^2 (2\pi)^4 \delta^{(4)} \left(p_A - \sum_{f=1}^{N} p_f \right) \tag{41}$$

s the differential decay rate (transition probability per unit time).

$$\Gamma = \int d\Gamma \tag{42}$$

a transition probability in unit time so

$$\frac{1}{\Gamma} \equiv \tau \tag{43}$$

is lifetime of the particle.

M, T matrices

It is important to remind that the $\mathcal{M}$ in the momentum space is equal:

$$i\mathcal{M}(2\pi)^4\delta^4\left(\sum_{i,f} p_i - p_f\right) = sum\, of\, all\, connected,\, amputated\, Feynman,\; diagrams. \tag{44}$$

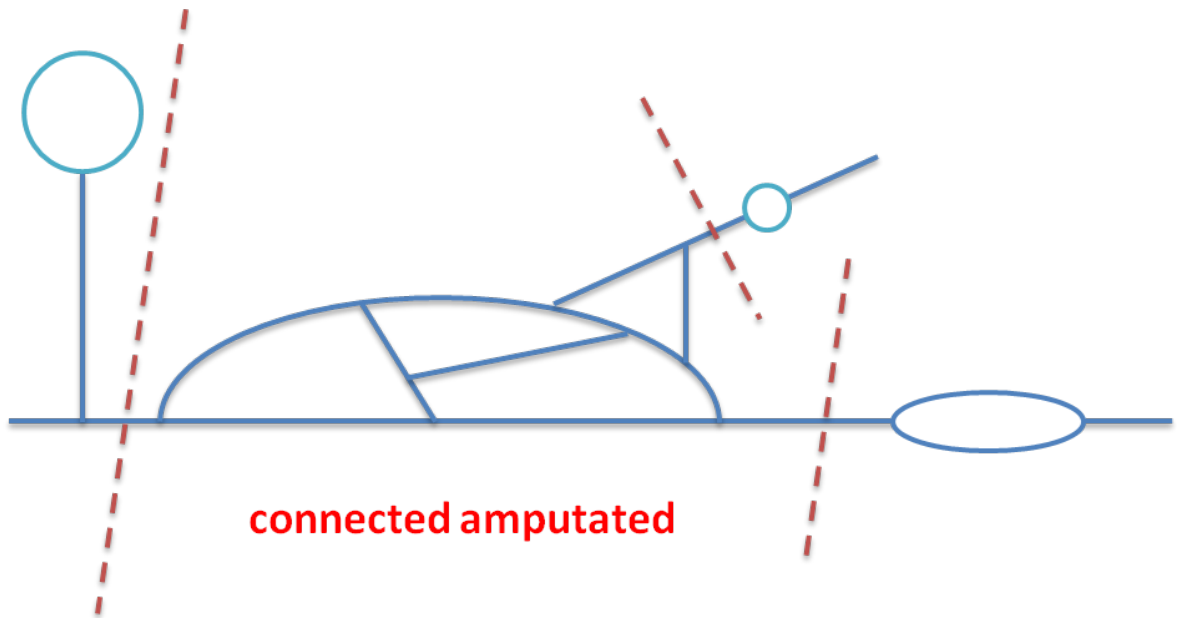
$\mathcal{M}$ transition amplitude is related with the T matrix:

$$\langle \vec{p}_1\vec{p}_2 ... \mid iT \mid \vec{k}_A\vec{k}_B\rangle = (2\pi)^4\delta^{(4)}\left(k_A + k_B - \sum p_f\right) i\mathcal{M}(k_A, k_B \to p_f). \tag{45}$$



T matrix element,

$$\langle \mathbf{p}_1 ... \mathbf{p}_n \mid iT \mid \mathbf{p}_A \mathbf{p}_B \rangle = \underbrace{\langle \mathbf{p}_1 ... \mathbf{p}_n \mid \sum_{N=1}^{\infty} \frac{1}{N!} \mathcal{T} \left[-i \int_{-t}^{t} dt' H_I(t') \right]^N \mid \mathbf{p}_A \mathbf{p}_B \rangle}_{connected\ amputated} \quad (46)$$



Units

Units

quantity	units typical for HEP	SI
length	1 fm	10^{-15} m
energy	1 GeV=10^9 eV	$1.602 \cdot 10^{-10}$ J
mass	1 GeV/c^2	$1.78 \cdot 10^{-27}$ kg
$\hbar = \frac{h}{2\pi}$	$6.582 \cdot 10^{-25}$ GeV·s	$1.055 \cdot 10^{-34}$ J s
c	$2.998 \cdot 10^{23}$ fm/s	$2.998 \cdot 10^{8}$ m/s
$\hbar c$	0.1973 GeV fm	$3.162 \cdot 10^{-26}$ J m

Heaviside-Lorentz units $\epsilon_0 = \mu_0 = \hbar = c = 1$ then $\alpha = e^2/4\pi = 1/137$

Remark

- Units can be expressed in terms of c and $\hbar$
- God given units (high energy units): $c = \hbar = 1$

$$\text{Energy} \quad : \quad \frac{\text{GeV}}{c^2} \rightarrow \text{GeV}$$

$$\text{Momentum} \quad : \quad \frac{\text{GeV}}{c} \rightarrow \text{GeV}$$

$$1\,\text{GeV}^{-1} = 0.1973\,\text{fm} \tag{47}$$

$$1\,\text{GeV}^{-1} = 6.59 \cdot 10^{-25}\,\text{s} \tag{48}$$

Units

Conversion Factors

- $$\sigma(\nu e \to \nu e) = G^2 s \tag{49}$$
- $G = [1/\text{energy}^2]$
- $s = [\text{energy}^2]$
- $\Rightarrow \sigma = [1/\text{energy}^2]$ but must be $[\text{length}^2]$
- notice that $\hbar c =$ energy length
- hence

$$\sigma(\nu e \to \nu e) = (\hbar c)^2 G^2 s$$

Scalar Field

Scalar Field

Lagrangian

Lagrangian for the scalar real field of mass m reads

$$\mathcal{L}_{scalar}(x) = \frac{1}{2}\partial_\mu\phi(x)\partial^\mu\phi(x) - \frac{m^2}{2}\phi^2(x) - \frac{\lambda}{4!}\phi^4, \quad 1 \gg \lambda > 0 \tag{50}$$

Remark

Notice that the Higgs boson is an example of the scalar field as well as the pions but the latter are described by the complex ($\pi^\pm$) and real (π^0) pseudoscalar fields.

After second quantization the scalar field operator reads

$$\phi(\mathbf{x},t)=\int\frac{d^3p}{\sqrt{2E_{\mathbf{p}}}(2\pi)^3}\left[a(\mathbf{p})e^{-iE_{\mathbf{p}}t+i\mathbf{p}\cdot\mathbf{x}}+a^{\dagger}(\mathbf{p})e^{iE_{\mathbf{p}}t-i\mathbf{p}\cdot\mathbf{x}}\right]. \tag{51}$$

or in the other form

$$\phi(\mathbf{x},t)=\int\frac{d^3p}{\sqrt{2E_{\mathbf{p}}}(2\pi)^3}\left[a(\mathbf{p})e^{-iE_{\mathbf{p}}t}+a^{\dagger}(-\mathbf{p})e^{iE_{\mathbf{p}}t}\right]e^{i\mathbf{p}\cdot\mathbf{x}}. \tag{52}$$

The annihilation and creation operators satisfy the commutation relation

$$[a(\mathbf{p}),a^{\dagger}(\mathbf{q})]=(2\pi)^3\delta^{(3)}(\mathbf{p}-\mathbf{q}) \tag{53}$$

One particle state is defined as,

$$\mid \mathbf{p}\rangle = \sqrt{2E_{\mathbf{p}}}a^{\dagger}(\mathbf{p}) \mid 0\rangle, \quad \text{where} \quad a(\mathbf{p}) \mid 0\rangle = 0. \tag{54}$$

Then

$$\langle \mathbf{q} \mid \mathbf{p}\rangle = 2E_{\mathbf{p}}(2\pi)^3\delta(\mathbf{p}-\mathbf{q}). \tag{55}$$

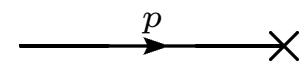
Notice that

$$\langle 0 \mid \phi(\mathbf{x},0) \mid \mathbf{p}\rangle = e^{i\mathbf{x}\cdot\mathbf{p}} \tag{56}$$

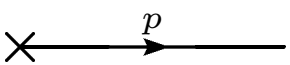
The Feynman propagator of the scalar field reads,

$$D_F(x-y) = \langle 0 \mid T(\phi(x)\phi(y)) \mid 0\rangle = \int \frac{d^4p}{(2\pi)^4}\frac{i}{p^2-m^2+i\epsilon}e^{-ip\cdot(x-y)}. \tag{57}$$

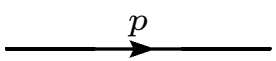
Feynman Rules



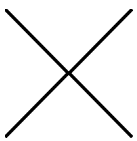
$= \quad 1$



$= \quad 1$



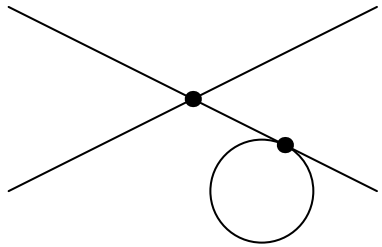
$$= \quad \frac{i}{p^2 - m^2 + i\epsilon}$$



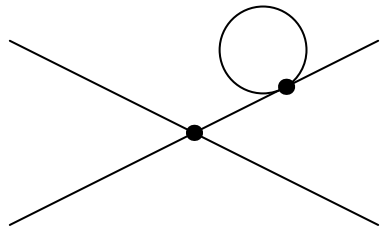
$= \quad -i\lambda$

- Impose the momentum conservation in each vertex;
- Integrate over each undetermined internal momentum $\int d^4l/(2\pi)^4$;
- Divide by the symmetry factor.

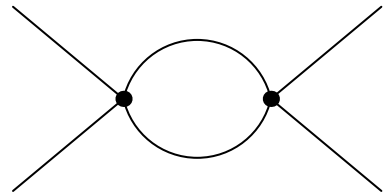
$2 \rightarrow 2$



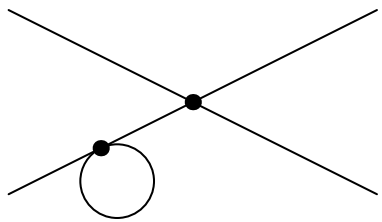
T1



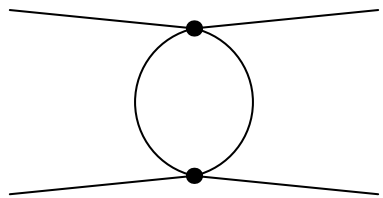
T2



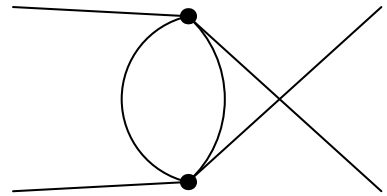
T3



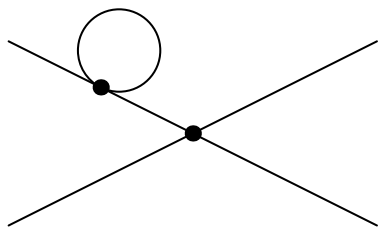
T4



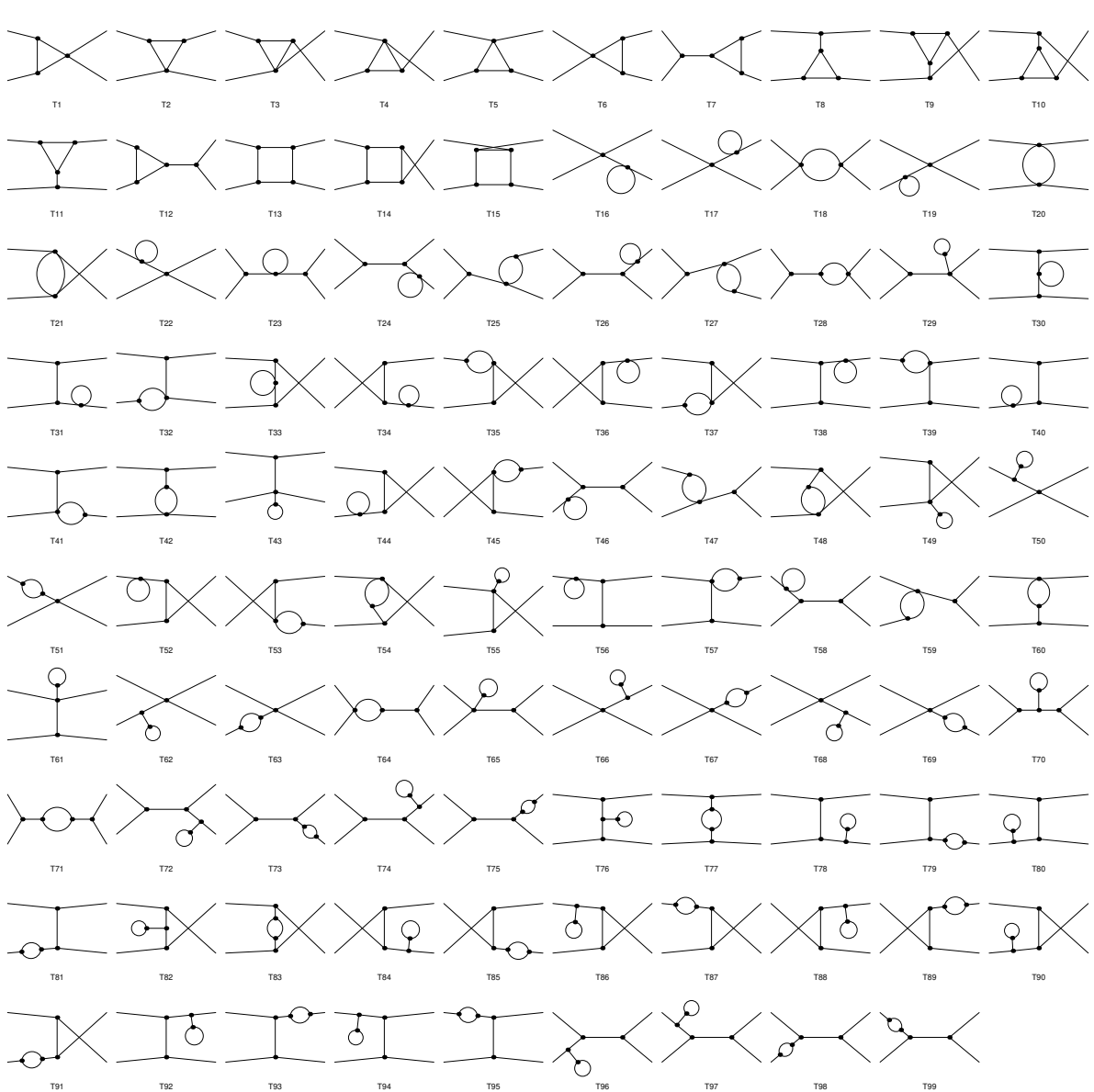
T5



T6



T7



generated with FeynArts, see also FeynCalc

Fermion field

Definition

Lagrangian for the Dirac spinor field

$$\mathcal{L}_{Dirac}(x) = \bar{\psi}(x)(i\gamma^{\mu}\partial_{\mu} - M)\psi(x). \tag{58}$$

The Dirac spinors satisfy the anti-commutation relation:

$$\{\psi_s(x), \psi_r^{\dagger}(y)\}\Big|_{x_0=y_0} = \delta^{(3)}(x-y)\delta_{sr} \tag{59}$$

and for the others,

$$\{\psi_s(x), \psi_r(y)\}|_{x_0=y_0} = 0, \quad \{\psi_s^{\dagger}(x), \psi_r^{\dagger}(y)\}\Big|_{x_0=y_0} = 0 \tag{60}$$

The spinor fields (in the Heisenberg picture) read

$$\psi(x) = \int \frac{d^3\mathbf{p}}{\sqrt{2E_p}(2\pi)^3} \sum_s \left(a(\mathbf{p},s)u(\mathbf{p},s)e^{-ip\cdot x} + b^\dagger(\mathbf{p},s)v(\mathbf{p},s)e^{ip\cdot x} \right), \quad (61)$$

$$\bar{\psi}(x) = \int \frac{d^3\mathbf{p}}{\sqrt{2E_p}(2\pi)^3} \sum_s \left(a^\dagger(\mathbf{p},s)\overline{u}(\mathbf{p},s)e^{ip\cdot x} + b(\mathbf{p},s)\overline{v}(\mathbf{p},s)e^{-ip\cdot x} \right), \quad (62)$$

The annihilation and creation operators satisfy the anti-commutation relations:

$$\{a(\mathbf{p},r), a^\dagger(\mathbf{q},s)\} = \{b(\mathbf{p},r), b^\dagger(\mathbf{q},s)\} = (2\pi)^3\delta^{(3)}(\mathbf{p}-\mathbf{q})\delta_{rs} \quad (63)$$

$$\{a(\mathbf{p},r), a(\mathbf{q},s)\} = \{b(\mathbf{p},r), b(\mathbf{q},s)\} = 0, \quad (64)$$

$$\{a^\dagger(\mathbf{p},r), a^\dagger(\mathbf{q},s)\} = \{b^\dagger(\mathbf{p},r), b^\dagger(\mathbf{q},s)\} = 0 \quad (65)$$

The wave functions u and v satisfy the Dirac equations in the momentum space:

$$(\hat{p} - M)u(\mathbf{p}, s) = 0, \tag{66}$$

$$(\hat{p} + M)v(\mathbf{p}, s) = 0, \quad \hat{p} \equiv \gamma_\mu p^\mu \tag{67}$$

We keep the normalization

$$u^\dagger(\mathbf{p}, s)u(\mathbf{p}, s') = 2E_{\mathbf{p}}\delta_{ss'}. \tag{68}$$

$$u(\mathbf{p}, s) = \sqrt{E_{\mathbf{p}} + M}\begin{pmatrix} \chi_s \\ \dfrac{\boldsymbol{\tau}\cdot\mathbf{p}}{E_{\mathbf{p}} + M}\chi_s \end{pmatrix}, \qquad v(\mathbf{p}, s) = \sqrt{E_{\mathbf{p}} + M}\begin{pmatrix} \dfrac{\boldsymbol{\tau}\cdot\mathbf{p}}{E_{\mathbf{p}} + M}\xi_s \\ \xi_s \end{pmatrix},$$

$$\chi_s^\dagger\chi_r = \delta_{rs} \qquad \xi_s^\dagger\xi_r = \delta_{rs}$$

where

$$\chi_{s=1/2} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \chi_{s=-1/2} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

where

$$\xi_{s=1/2} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \xi_{s=-1/2} = -\begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Notice that

$$\xi_s = -i\tau^2\chi_s = (-1)^{(1-2s)/2}\chi_{-s} \tag{69}$$

$$\overline{u}(\mathbf{p}, s)u(\mathbf{p}, s) = 2M \tag{70}$$

$$\sum_s u(\mathbf{p}, s)\overline{u}(\mathbf{p}, s) = \hat{p} + M, \quad \hat{p} = \gamma \cdot p \tag{71}$$

Definition

One particle state is defined as:

$$|\mathbf{p}, s\rangle \equiv \sqrt{2E_{\mathbf{p}}} a_{\mathbf{p}}^{s\dagger} |0\rangle \tag{72}$$

Therefore

$$\langle \mathbf{q}, r \ |\mathbf{p}, s\rangle = 2E_p (2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{q}) \delta_{rs} \tag{73}$$

Notice that,

$$a(\mathbf{p}, s) \mid 0\rangle = b(\mathbf{p}, s) \mid 0\rangle = 0. \tag{74}$$

The Feynman propagator

$$[S_F(x-y)]_{ab} = \langle 0 \mid \mathrm{T}\psi_a(x)\bar{\psi}_b(y) \mid 0\rangle = \frac{i}{(2\pi)^4} \int d^4p e^{-ip\cdot(x-y)} \frac{(\hat{p} + M)_{ab}}{p^2 - M^2 + i\epsilon} \tag{75}$$

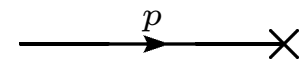
Feynman rules for $i\mathcal{M}$ in QED

QED Lagrangian

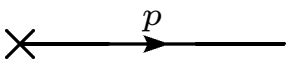
$$\mathcal{L}_{QED}(x) = \underbrace{\bar{\psi}(x)(i\gamma^\mu \partial_\mu - m)\psi(x)}_{Dirac} - \underbrace{\frac{1}{4}F_{\mu\nu}(x)F^{\mu\nu}(x)}_{Maxwell} \underbrace{-e\bar{\psi}(x)\gamma^\mu\psi(x)A_\mu(x)}_{Interaction}, \quad (76)$$

$$F_{\mu\nu}(x) = \partial_\mu A_\nu(x) - \partial_\nu A_\mu(x) \quad (77)$$

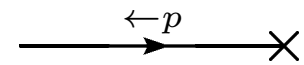
- We keep the normalization $e = -|e|$. For a fermion of charge $Q|e|$ ($Q > 0$), replace $e \to -Qe$, i.e. the vertex reads $+iQe\gamma^\mu$.



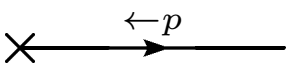
$= \quad u(\mathbf{p}, s)$



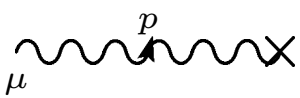
$= \quad \bar{u}(\mathbf{p}, s)$



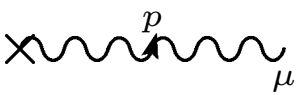
$= \quad \overline{v}(\mathbf{p}, s)$



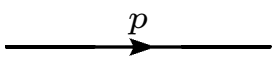
$= \quad v(\mathbf{p}, s)$



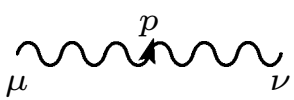
$= \quad \epsilon^{\mu}(p, \lambda)$



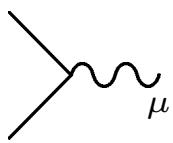
$= \quad \epsilon^{\mu *}(p, \lambda)$



$= \quad \dfrac{i(\hat{p} + m)}{p^2 - m^2 + i\epsilon}$



$= \quad -\dfrac{ig_{\mu\nu}}{p^2 + i\epsilon}$



$= \quad -ie\gamma^{\mu} \quad \text{or} \quad iQe\gamma^{\mu}$

Diagrams drawn with use of feyn package

- Impose the momentum conservation in each vertex;
- Integrate over each undetermined internal momentum $\int d^4l/(2\pi)^4$;
- Figure out the overall sign of the diagram;
- Relative factor (-1) appears for the diagrams that differ only by interchange of two external fermion lines;
- Insert the additional factor -1 for each closed fermion loop.

What have we obtained?

$$d\sigma, \quad d\Gamma, \quad S \to T \to \mathcal{M}$$